

BRAMON, Sweden

WMO: 023680

Lat: 62.220N

Lon: 17.739E

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.2	-12.0	-16.6	0.9	-13.3	-14.5	1.1	-11.1	16.6	-2.0	15.5	-2.6	5.0	320	0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	22.1	18.0	20.7	17.2	19.4	16.3	19.1	21.1	18.0	19.9	16.9	18.7	4.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	20.4	17.1	12.2	19.2	16.1	11.4	18.0	54.5	21.2	50.8	20.0	47.4	18.8	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.7	11.2	-16.2	25.2	3.7	2.3	-18.8	26.9	-21.0	28.2	-23.0	29.5	-25.7	31.2		
(5)			-16.9	19.3	3.7	1.3	-19.6	20.2	-21.8	21.0	-23.9	21.7	-26.6	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-2.4	-2.6	-0.3	3.3	7.8	12.6	16.1	15.6	11.7	6.4	2.2	-0.7	(6)
(7)		DBStd	7.42	4.43	4.85	4.03	2.85	2.87	2.62	2.41	2.26	2.57	2.84	3.25	4.23	(7)
(8)		HDD10.0	2041	385	353	320	202	82	6	0	0	13	116	234	330	(8)
(9)		HDD18.3	4569	643	587	578	452	328	172	78	89	201	370	483	588	(9)
(10)		CDD10.0	528	0	0	0	1	12	84	191	174	63	4	0	0	(10)
(11)		CDD18.3	15	0	0	0	0	0	1	9	5	0	0	0	0	(11)
(12)		CDH23.3	17	0	0	0	0	1	1	13	2	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	6.0	5.8	5.5	4.9	4.4	4.7	4.5	4.7	4.9	5.7	5.8	5.7	(14)
(15)	Precipitation	PrecAvg	544	32	24	26	32	40	49	56	70	53	60	56	41	(15)
(16)		PrecMax	842	74	42	61	80	77	78	165	134	125	154	160	85	(16)
(17)		PrecMin	400	3	8	6	6	6	6	10	16	16	6	15	2	(17)
(18)		PrecStd	103	19	10	14	22	21	21	35	33	25	42	36	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	8.1	10.7	14.2	18.0	21.8	24.8	23.2	19.5	14.1	10.8	8.3	(19)
(20)			MCWB	4.2	4.5	6.2	9.0	11.2	14.7	18.5	18.9	14.7	11.0	7.9	6.1	(20)
(21)		2%	DB	5.1	6.2	8.4	11.6	15.5	19.7	22.7	21.6	17.4	12.2	8.5	6.7	(21)
(22)			MCWB	3.0	3.4	4.6	7.1	10.7	14.3	18.4	18.3	14.2	10.3	6.9	4.9	(22)
(23)		5%	DB	4.0	4.7	6.7	9.4	13.7	18.0	21.3	20.2	16.2	11.2	7.2	5.5	(23)
(24)			MCWB	2.1	2.4	3.6	5.7	9.5	13.4	18.0	17.3	13.5	9.7	6.1	3.9	(24)
(25)		10%	DB	3.0	2.9	5.0	7.6	12.0	16.5	19.9	19.0	15.1	10.3	6.2	4.5	(25)
(26)			MCWB	1.5	1.2	2.4	4.7	8.5	12.8	17.1	16.5	12.9	8.8	5.3	3.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.8	6.6	9.1	12.6	16.2	20.9	20.6	16.0	11.9	8.2	6.7	(27)
(28)			MCDWB	6.3	7.4	10.2	13.6	16.3	19.4	22.5	22.1	17.8	12.7	9.7	7.7	(28)
(29)		2%	WB	3.4	3.6	5.0	7.4	11.1	15.1	19.5	19.0	15.0	11.0	7.2	5.2	(29)
(30)			MCDWB	4.8	5.8	7.9	11.0	14.7	18.4	21.6	20.8	16.5	11.9	8.2	6.3	(30)
(31)		5%	WB	2.4	2.5	3.7	6.1	10.0	14.2	18.4	18.0	14.2	10.0	6.4	4.2	(31)
(32)			MCDWB	3.7	4.2	6.3	8.9	13.0	17.0	20.6	19.6	15.6	11.1	7.0	5.2	(32)
(33)		10%	WB	1.6	1.5	2.6	5.0	8.9	13.5	17.3	17.0	13.4	9.0	5.6	3.3	(33)
(34)			MCDWB	2.7	2.7	4.7	7.3	11.5	15.8	19.4	18.4	14.8	10.0	6.2	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.0	4.2	4.6	4.6	4.7	4.9	4.2	4.3	4.2	3.6	3.1	3.9	(35)
(36)			MCDBR	4.6	5.4	7.0	8.2	7.8	7.0	6.0	5.4	5.6	4.5	4.0	4.4	(36)
(37)			MCWBR	3.8	4.1	4.6	4.8	4.4	3.6	3.4	3.4	3.6	3.5	3.5	3.7	(37)
(38)		5% WB	MCDWB	4.1	5.1	6.5	7.8	7.1	5.8	4.7	4.4	4.9	3.8	3.5	4.0	(38)
(39)			MCWBR	3.6	4.0	4.5	4.9	4.4	3.5	3.2	3.1	3.6	3.5	3.3	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.244	0.299	0.319	0.356	0.356	0.351	0.370	0.362	0.330	0.316	0.262	0.344	(40)	
(41)		taud	2.140	2.268	2.364	2.378	2.443	2.498	2.464	2.478	2.550	2.442	2.190	2.598	(41)	
(42)		Ebn at Noon	442	660	794	832	864	875	845	820	787	646	409	275	(42)	
(43)		Edh at Noon	33	61	80	97	98	95	96	87	68	52	31	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.20	0.81	2.31	3.82	4.98	5.71	5.25	3.99	2.48	1.08	0.30	0.11	(44)	
(45)		RadStd	0.03	0.14	0.26	0.34	0.50	0.36	0.54	0.33	0.29	0.19	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-105	N/A	N/A		

Nomenclature: See separate page